Evidence first.
Curiosity always.

A connected portfolio of reproducible neuroscience experiments, applied neurotechnology, and communication artifacts.

PROJECT	VERIFIED STATUS	PRIMARY EVIDENCE
Mouse visual cortex	Three-mouse replication complete; population extension in progress	Sensorium report, replication summary, figures
EEG eye state	Time-aware study plus label-shuffle control complete	Pipeline, analysis, metrics, report, explainer
Neurodivergent / Neurocomm	Applied project references and templates collected	Mobile-first HTML, presentation, PDF, video patterns

1 / Portfolio thesis

We study how neural activity relates to behavior, then translate the work into accessible tools and explanations. The portfolio moves from a scientific question to a public, reviewable artifact without hiding uncertainty or collaboration boundaries.

2 / Flagship study

The mouse visual-cortex project compares stimulus-only and behavior-augmented ridge-regression encoding models on Sensorium recordings. Under the same pipeline, the original mouse rises from 0.1265 to 0.1583, a second mouse rises from 0.1543 to 0.2034, and a third rises from 0.1431 to 0.1905. Pupil improves all three animals; running varies by animal. A leave-one-animal-out sensitivity check keeps the mean gain positive in every omission. The claim remains about prediction, not causation.

3 / Complementary study

The EEG eye-state project tests whether causal temporal context improves classification from a public 14-channel EEG recording. The pointwise baseline reaches test ROC AUC 0.625; the first context extension reaches 0.530 on the final chronological block. Five forward-chaining folds average 0.478 and 0.488 ROC AUC, while 20 global label shuffles average 0.499 and 0.501. A 90-sample purge-gap sensitivity analysis averages 0.448 and 0.443. The negative result is retained because validation improvement did not guarantee held-out generalization.

4 / Translation layer

The neuro-divergent and neurocomms work contributes human-centered patterns for communication and accessibility applications. Those patterns are reused as mobile specifications, HTML explainers, presentation surfaces, PDF examples, video scripts, and scope/limitations notes for the research projects.

5 / Evidence and reproducibility

The umbrella repository contains the reviewer index, contribution ledger, reproducibility checklist, interactive explainers, project reports, presentation outlines, and video production gates. The flagship repository now includes a core dependency manifest, a written replication protocol, and behavior-shuffle controls across three animals. The EEG project retains the downloaded ARFF data, provenance note, analysis script, metrics JSON, label-shuffle control, and report.

6 / Honest remaining work

Before submission, the applicant must personalize the essays with genuine experiences, extend the mouse replication beyond three animals for population-level uncertainty, and record the final videos in the applicant’s own voice. Descriptive within-animal bootstrap intervals are now documented, but this packet does not present population-level uncertainty as complete.

Portfolio links

GitHub hub: github.com/mehtama1234/ASDRP-Neuroscience-Portfolio
Flagship code: github.com/AmishI123/neuroscience-project
Live portfolio: asdrp-neuroscience-portfolio.gargibhatia37240.chatgpt.site

Generated from the committed application materials. Source Markdown remains alongside this PDF.